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Seaweed extracts elicitation improves leaf and essential oil production on *Pelargonium graveolens* L. Herit

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ABSTRACT

Little is known about the role of seaweed extracts in modulating the growth and the essential oil production in geranium [*Pelargonium graveolens* (L.) Herit]. Hence, this work aimed to evaluate the effects of *Ascophylum nodosum* (AN) and *Solieria chordalis* (SC) extracts on biometric, biochemical and phytochemical parameters in geranium plants. The experiment was conducted at greenhouse conditions, and seaweed extracts were applied by foliar pulverization. The obtained results revealed that seaweed extracts enhanced vegetative growth, directing it towards the production of a greater number of leaves at the expense of plant height and ramifications. Additionally, chlorophyll *b*, total soluble sugars and essential oil yield were increased due to seaweed extracts. The main constituents of volatile oil such as Linalool, Citronellol and *iso*-Menthone were not changed by seaweed extract treatments, with exception of Geraniol content which were slight reduced in SC treated plants. These results suggested that seaweed extracts may be used as an eco-friendly alternative to modulate growth and productivity of geranium.

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Aromatic plant; biostimulant; plant metabolism; sustainable production

1. Introduction

Geranium [*Pelargonium graveolens* (L.) Herit] is an aromatic species that produces essential oil with great commercial potential, as its oil is sold to different industries, for example cosmetic, perfumery, food and beverages, veterinary medications and medicine (1,2). Additionally, geranium essential oil also has therapeutic actions in the treatment of menopause and anxiety symptoms (3). The leaves have glandular trichomes, which store the essential oil with a unique rose aroma, due mainly to the presence of the compounds Citronellol and Geraniol (4). Nonetheless, the essential oil content in geranium leaves is quite low when compared to other aromatic plants (around 0.08% to 1%), causing this species to have a low distillation yield and a high essential oil price on the market.

Agronomic studies on geranium have been focused on aspects related to production management (planting spacing, climatic effects, harvest times, forms of fertilization and production systems). Correct crop management aims to obtain a good production of leaf biomass, increasing the extraction yield (amount of essential oil produced per plant and per hectare). In geranium, essential oil synthesis occurs in both adult photosynthetic leaves and young leaves that are expanding rapidly. The latter require a greater supply of energy and photoassimilates for their development (5). In this species, the high photosynthetic activity was related to the greater fresh and dry mass of the leaves and the essential oil content in them (1,6).

Biostimulants promote greater plant growth and productivity via activation of different physiological and biochemical processes (7). By definition, biostimulants are compounds of biological origin or microorganisms that, when applied to plants, result in greater productivity and greater resilience to stress (8). Products based on seaweed extracts are plant biostimulants, whose plant action mechanisms are attributed to a rich mix of components such as carbohydrates, plant hormones, phenolics, nutrients and polysaccharides (9,10). In the case of aromatic and medicinal species, the application of biostimulants can also result in an increase in the concentration of bioactive substances present in the plant, due to the activation of genes related to the biosynthetic pathways that produce these substances (11). The stimulation of the secondary metabolites in plants is a technique called elicitation, which can be performed both in vitro and in vivo (12). The use of seaweed extracts as plant elicitors has been successfully tested in different medicinal and aromatic plants. Among the seaweed species with documented biostimulatory

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activities on major crops, the species *Ascophyllum nodosum* is one of the most important (10). Recently, the red seaweed *Solieria chordalis* (C. Agardh) J. Agardh 1842 has been used in agriculture as extract to increase crop yields (13,14). As far as it is known, the influence of these algae species on *Pelargonium graveolens* is poorly studied. In this way, the current study was designed to examine the effects of *A. nodosum* and *S. chordalis* on growth, biochemical traits and essential oil production of geranium plants.

2. Material and methods

2.1. Greenhouse conditions

The experiment was carried out under greenhouse conditions. *Pelargonium graveolens* plant material was obtained in the selection and breeding program from the Agronomic Institute of Campinas (IAC), Campinas, São Paulo, Brazil. In order to homogenize the material, the seedlings were selected with similar height (around 15 cm) and number of leaves (approximately five fully expanded leaves). The seedlings were planted in pots with a capacity of 7 liters, filled with a mixture of commercial vegetable soil and white sand (6:4), which had the following chemical characteristics: organic matter, 68.5 g dm^{-3}, pH (1:2.5 soil/suspension of $CaCl_2$ 0.01 mol L^{-1}), 6.2, P (resin), 305 mg dm^{-3}, K, Ca and Mg exchangeable from 10.6, 193 and 45.8 $mmol_c$ dm^{-3}, respectively, total acidity in pH 7.0 (H+Al) of 17.2 $mmol_c$ dm^{-3}, total cation exchange capacity (CEC) of 267.5 $mmol_c$ dm^{-3} and base saturation of 93%. The plants were irrigated frequently (every 1–2 days) to keep soil moisture at field capacity (close to 100%).

The elicitors were applied by foliar spraying with at the following five application timings: 45, 60, 75, 90 and 105 days after transplanting – DAT. Seaweed extracts were tested with *Ascophyllum nodosum* [commercial product Acadian (Acadian Seaplants, Dartmouth, Canada)] and *Solieria chordalis* [commercial product Seamel Pure (Olmix, Brehan, France)] at a dose of 5 mL L^{-1} and 2 mL L^{-1}, respectively. The concentration of seaweed extracts were selected based on literature data. The volume of solution applied per plant was estimated by previous spraying tests, based on the drop point, with the application of increasing volumes (mL) according to the increase in height and ramifications of the plants (20 mL per plant at 45 and 60 DAT, and 45 mL per plant at 75, 90 and 105 DAT). A manual spray was used, adding Agral® [nonyl phenoxy poly-(ethyleneoxy) ethanol] as an adhesive spreading agent in the proportion of 50 μL L^{-1} of solution, in the case of treatments with seaweed extracts. Control plants were sprayed only with tap water.

2.2. Biomass yield and biochemical analyses

Biometric evaluations were performed at 130 DAT with determinations of plant height (cm), number of branches per plant, number of leaves per plant, leaf fresh mass (g plant $^{-1}$), branches fresh mass (g plant $^{-1}$) and total fresh mass (leaves + branches). The measurements were done in fresh material like reported in previous studies with *P. graveolens* (1), considering that producers use the fresh plant to extract the essential oil. Furthermore, leaf fresh mass might be more able to reflect the physiological functions of leaves associated with photosynthesis and respiration than leaf dry mass (15). For the biochemical analysis, the third leaf completely expanded from the apex of the stem was collected, with immediate freezing at −80°C for later determination of the concentration (μg/g) of chlorophylls *a* and *b*, carotenoids and anthocyanin by extraction with acetone 80% (16), and the content of total soluble carbohydrates (mg/g) by extraction with phenol and sulfuric acid and glucose standard curve (17).

2.3. Essential oil extraction and analysis

The essential oil (EO) content of the leaves was determined in the fresh leaves, through hydrodistillation in a Clevenger-type apparatus. Samples of 150 g of leaves per repetition were ground together with 1200 mL of distilled water in a blender. The mixture was transferred to a 2000 mL volumetric flask and extracted for 2.5 h. At the end of the extraction period, the essential oil was collected through a phase separation and weighed on an analytical balance to determine the mass (g) of oil. The percentage content and yield of essential oil in the leaves were determined according to Equations 1 and 2:

$$\text{EO content}(\%) = \left(\frac{\text{essential oil mass(g)}}{150(\text{g})}\right) \times 100 \quad (1)$$

$$\text{EO yield}(\text{g plant}^{-1}) = \frac{\text{leave fresh mass(g plant} - 1) \times \text{ OE(g)}}{150(\text{g})} \quad (2)$$

The extracted oil was stored in a freezer at −20°C until analysis. In order for the identification and quantification of the essential oil components, the samples were diluted in hexane until an oil concentration of 1%. A 1.0 μL aliquot of such solution was injected into a gas chromatograph coupled with a mass spectrometer (GC/MS) (Shimadzu 2010 Plus) – UFPR Chemistry Department, using an injector maintained at 250°C. Separation of the constituents was performed using an HP-5 MS capillary column (30 m x 0.25 mm x 0.25 µm) with helium gas as a carrier (1.0 mL.min^{-1}). The oven temperature was programmed to gradually increase from 60°C to 240°C at a rate of 3°C.min^{-1}. Chemical constituent identification was conducted by comparing their mass spectra with a database, and also their linear retention indices, calculated by the injection of a homologous series of *n*-alkanes (18), which were compared with data available in the literature (19). Meanwhile, for compound quantification, a GC coupled with a flame ionization detector (FID) was used, in the same conditions described above, with the exception of the carrier gas (hydrogen, at 1.5 mL.min^{-1}). The percentage composition, in turn, was obtained by the electronic integration of the FID signal, by dividing the area of each component by the total area (%).

The experimental design adopted was completely randomized, with three treatments (Control, SC = *S. chordalis* extract and AN = *A. nodosum* extract) and eight replications. Each replication was composed of three plants, totalizing seventy-two plants, with the mean of the three plants being determined. The data obtained were tested for normality and homogeneity of variance and submitted to analysis of variance ($p \leq 0.05$). Subsequently, the means were compared using the Duncan's Test ($p < 0.05$). The data on the chemical composition of the essential oil was analyzed using principal component analysis (PCA) with the Statistica 7 program (StatSoft, 2020).

3. Results

3.1. Biometric and biochemical data

The plants treated with seaweed extracts showed non-significant difference in height, number of branches and fresh mass of branches (Figure 1a–c). However, both seaweed extracts resulted in increased number of leaves per plant (12.98% for SC and 14.08% for AN treatments, respectively) (Figure 1b). Plants treated with SC extract showed higher leave fresh mass (17.66%) and total fresh mass (13.75%) when compared to the control plants (Figure 2a–c).

The SC- and AN-treated plants showed reductions in chlorophyll *a* and increases in chlorophyll *b* in relation to the control (Figures 3a,b). The plants treated with both seaweed extracts did not show variations in concentrations of carotenoids and anthocyanins compared

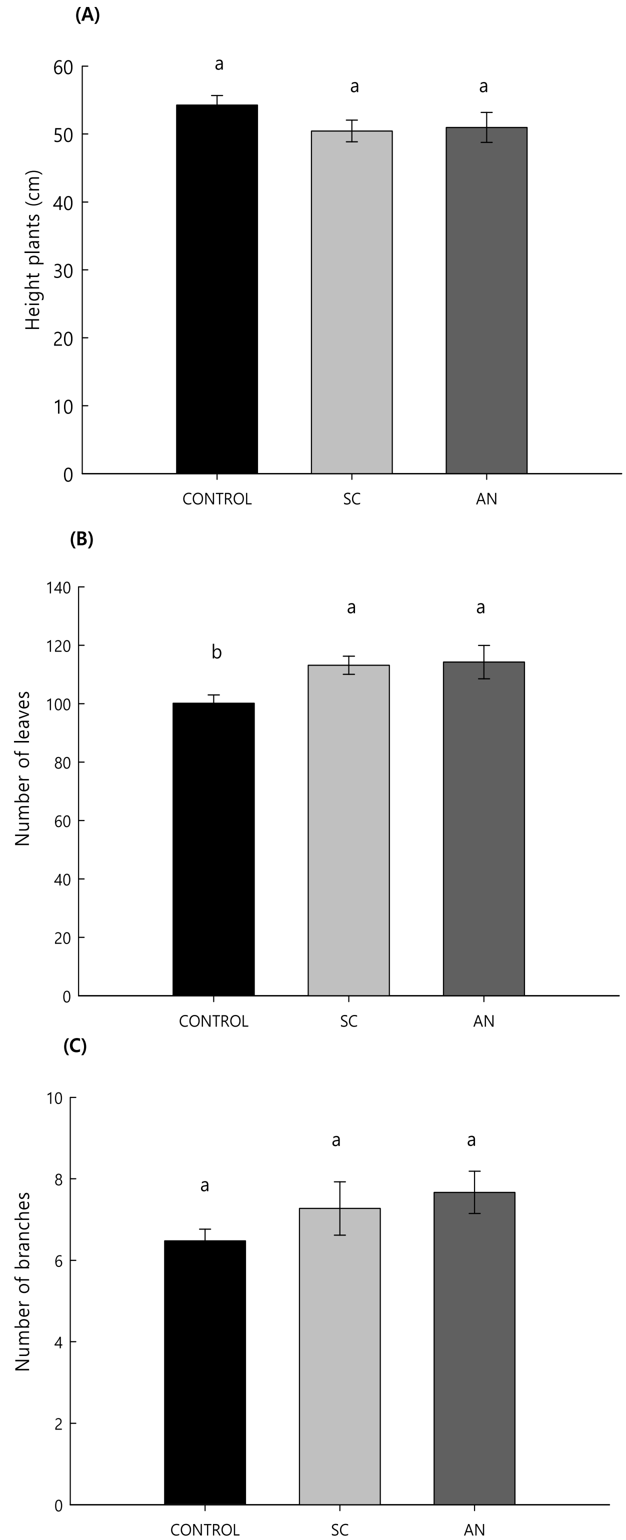

Figure 1. Height, Number of Leaves and Number of Branches in *Pelargonium Graveolens* Plants Treated with *Solieria Chordalis* (SC) and *Ascophylum Nodosum* (AN) Extracts. Different Letters Indicate Significant Differences According to Duncan's Test ($p < 0.05$). Bars Represent the Standard Error ($N = 8$).

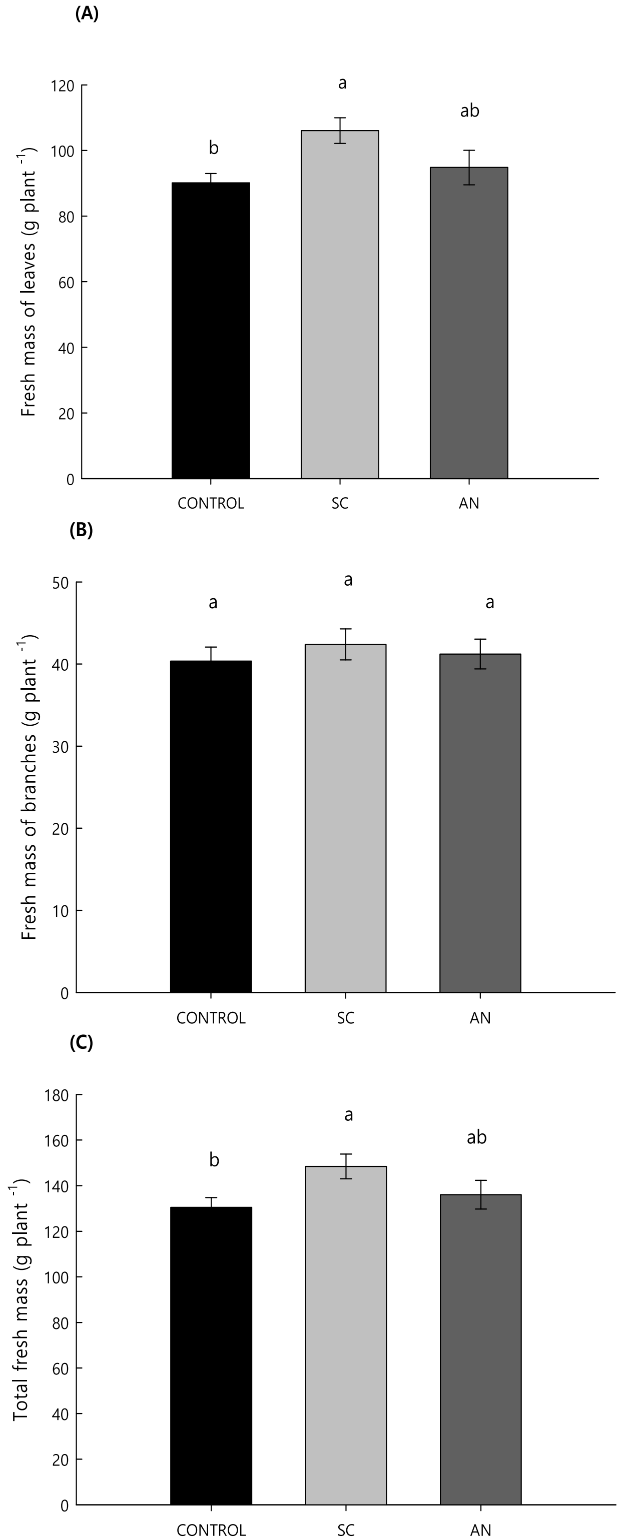

Figure 2. Leaf Fresh Mass, Branches Fresh Mass and Total Fresh Mass in *Pelargonium Graveolens* Plants Treated with *Solieria Chordalis* (SC) and *Ascophylum Nodosum* (AN) Extracts. Different Letters Indicate Significant Differences According to Duncan's Test ($p < 0.05$). Bars Represent the Standard Error ($N = 8$).

to the control (Figures 3d,e). Plants treated with SC and AN extracts showed higher concentrations of total soluble sugars in the leaves, with increases of 27.56% and 66.6%, respectively, in relation to the control (Figure 3f).

3.2. Concentration and chemical composition of essential oil

Regarding the concentration of EO (%) present in leaves, there was a significant increase of 27.36% in the plants treated with AN compared to the control (Figure 4a). Both seaweed extracts resulted in higher essential oil yield per plant, with increments of 42% for SC treatment and 50.2% for AN treatment (Figure 4b).

The compounds identified in the EO and their average relative percentages in the samples are presented in Table 1 and the chromatogram is presented in Figure 5. A mean of 91.96% of the compound composition was identified, with twenty-four substances identified. The substances with the highest relative proportion were Geraniol, Citronellol, Linalool and *iso*-Menthone (Table 1 and Figure 5) with means of 29.22%, 15.77%, 14.60% and 8.82%, respectively, among the treatments. These substances have been reported as majority in specimens of *P. graveolens* (20,21). Plants treated with SC presented a slight reduction (10.82%) in the geraniol content compared to control (Table 2). Principal component analysis explained 99.31% of the variation in essential oil compounds, with the most of this variation presented in the horizontal axis (Figure 6). The most prominent volatile compound, present in all treatments, was Geraniol, followed by the compounds Linalool and Citronellol. These compounds formed separated groups whose were correlated to the effect of the SC and AN treatments.

4. Discussion

Aromatic plants that are sources of essential oils growing in optimal conditions usually accumulate small amounts of this raw material. Elicitation is an efficient method used to stimulate plant metabolism, inducing an elevation in the content of essential oil (8,14,15). According to hormesis phenomenon, the eustress-induced plant defense response by elicitors enhances the plant secondary metabolism (22). Given the low essential oil yield in *Pelargonium graveolens* and the commercial importance of its essential oil, the effective-

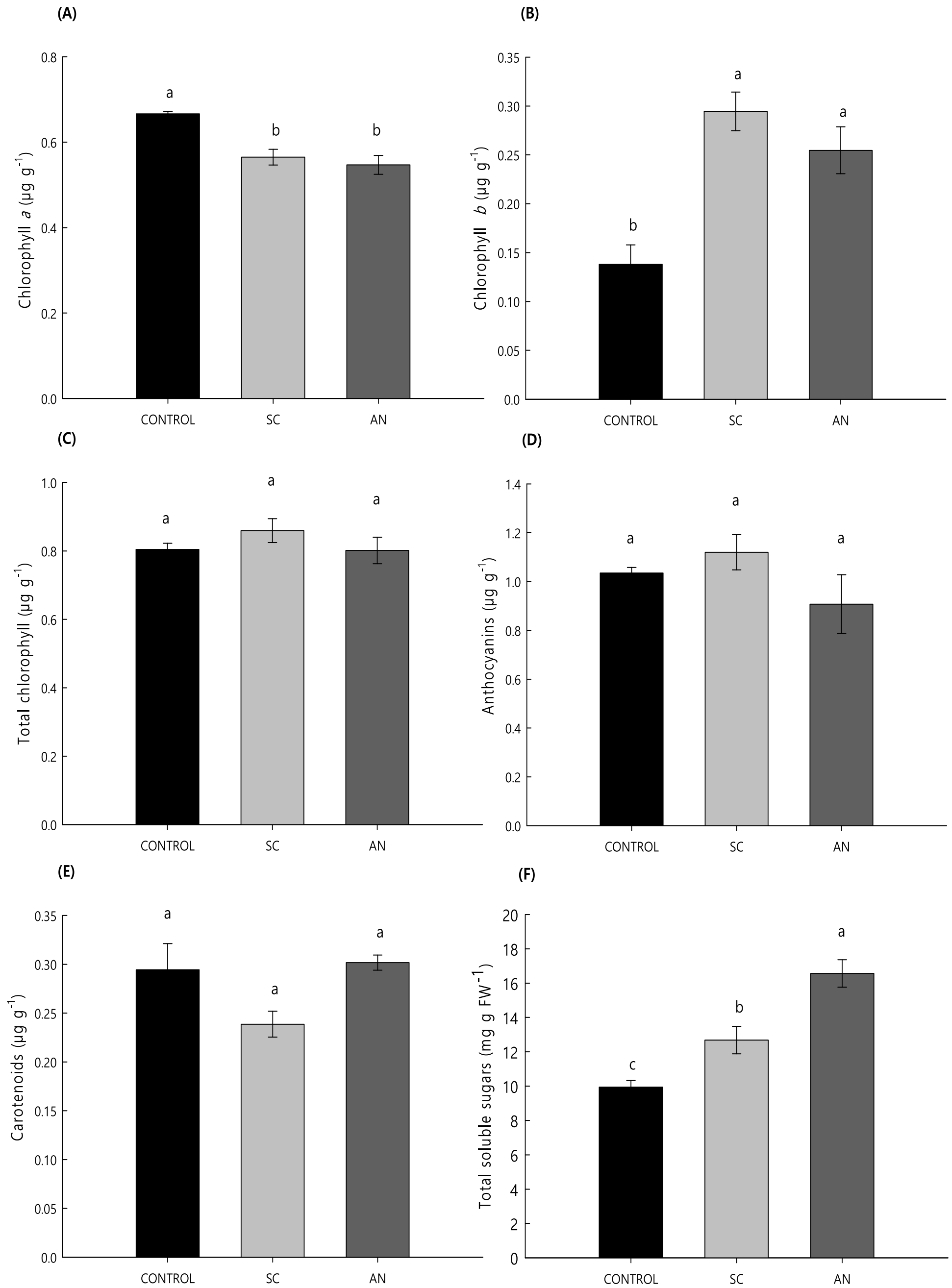

Figure 3. Foliar Concentration of Chlorophyll *a*, Chlorophyll *B*, Total Chlorophylls, Carotenoids, Anthocyanins and Total Soluble Sugars in *Pelargonium Graveolens* Plants Treated with *Solieria Chordalis* (SC) and *Ascophylum Nodosum* (AN) Extracts. Different Letters Indicate Significant Differences According to Duncan's Test ($p < 0.05$). Bars Represent the Standard Error ($N = 8$).

ness of seaweed extracts in the elicitation of this species was assessed in a pot cultivation experiment.

Seaweed biostimulants based on *Ascophylum nodosum* and *Solieria chordalis* were shown to be effective in geranium plants, where foliar applications enhanced vegetative growth, directing it towards the production of a greater number of leaves at the expense of plant height and the number of branches on the plant (Figures 1 and 2). This result is interesting especially in *Pelargonium* species where it has been shown that, compared to branches, leaves contain the majority of the essential oil (21). Furthermore, glandular trichomes density in the young leaves of cultivated *Pelargonium* species have also been correlated with essential oil yields (20).

The seaweed extracts tested also promoted an increase in the concentrations of chlorophyll *b* and carbohydrates (Figure 3B) in the leaves of geranium, being

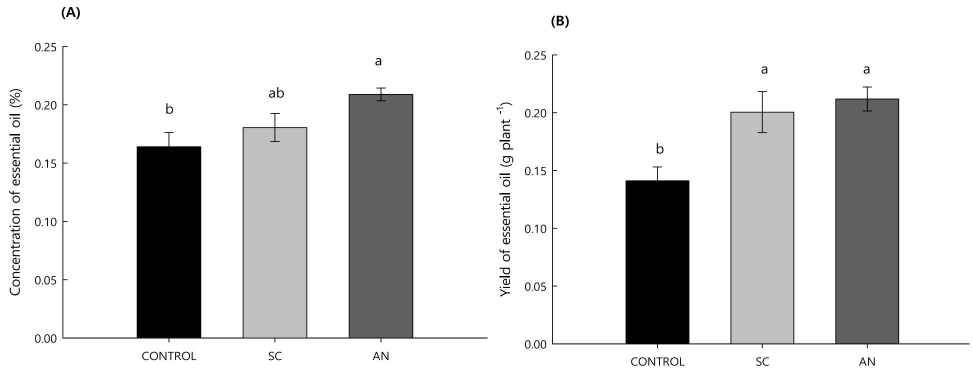

Figure 4. Essential Oil Content (A) and Yield (B) in *Pelargonium Graveolens* Plants Treated with *Solieria Chordalis* (SC) and *Ascophylum Nodosum* (AN) Extracts. Different Letters Indicate Significant Differences According to Duncan's Test ($p < 0.05$). Bars Represent the Standard Error ($N = 8$).

Table 1. Average Chemical Composition (%) of the Essential Oil of *Pelargonium Graveolens* in Response to Seaweed Treatments (SC = *Solieria Chordalis* Extract, an = *Ascophyllum Nodosum* Extract).

	Compounds	RIc	RII	% (±sd)		
				Control	SC	AN
1	Myrcene	989	988	0.28 (±0.03)	0.20 (±0.01)	0.31 (±0.03)
2	**Linalool**	**1100**	**1095**	**14.13 (±2.84)**	**16.58 (±1.39)**	**13.07 (±0.73)**
3	*cis*-Rose oxide	1109	1106	0.21 (±0.05)	0.18 (±0.03)	0.21 (±0.02)
4	Menthone	1150	1148	0.29 (±0.06)	0.30 (±0.01)	0.28 (±0.01)
5	***iso*-Menthone**	**1161**	**1158**	**9.22 (±1.88)**	**8.73 (±0.53)**	**8.52 (±0.94)**
6	*iso*-Menthol	1179	1179	0.53 (±0.06)	0.42 (±0.05)	0.53 (±0.03)
7	α-Terpineol	1187	1186	0.59 (±0.20)	0.82 (±0.12)	0.55 (±0.09)
8	**Citronellol**	**1229**	**1223**	**16.03 (±1.53)**	**14.83 (±0.87)**	**16.45 (±0.34)**
9	Neral	1239	1235	0.41 (±0.07)	0.36 (±0.03)	0.35 (±0.04)
10	**Geraniol**	**1257**	**1249**	**31.13 (±1.53)**	**27.76 (±1.51)**	**28.78 (±1.27)**
11	Geranial	1269	1264	1.28 (±0.47)	1.18 (±0.17)	0.92 (±0.14)
12	α-Copaene	1370	1374	0.89 (±0.12)	0.88 (±0.08)	0.93 (±0.08)
13	β-Bourbonene	1378	1387	3.99 (±1.05)	4.03 (±0.78)	2.36 (±0.24)
14	(*E*)-Caryophyllene	1412	1417	0.59 (±0.25)	0.54 (±0.26)	0.46 (±0.04)
15	Citonellyl propanoate	1438	1444	3.52 (±0.44)	4.06 (±0.33)	5.03 (±0.46)
16	*cis*-Muurola-3,5-diene	1442	1448	0.33 (±0.04)	0.37 (±0.04)	0.43 (±0.03)
17	Germacrene D	1474	1480	1.81 (±0.08)	2.00 (±0.14)	2.51 (±0.22)
18	Bicyclogermacrene	1494	1500	-	0.13 (±0.02)	0.16 (±0.03)
19	δ-Cadinene	1518	1522	0.27 (±0.04)	0.31 (±0.03)	0.39 (±0.05)
20	Geranyl butanoate	1560	1562	0.71 (±0.14)	0.74 (±0.09)	0.79 (±0.04)
21	2-Phenyl ethyl tiglate	1581	1584	1.40 (±0.27)	1.32 (±0.11)	1.57 (±0.09)
22	10-*epi*-γ-Eudesmol	1610	1622	3.59 (±0.92)	3.47 (±0.29)	3.87 (±0.30)
23	Citronellyl tiglate	1664	1666	0.16 (±0.05)	0.17 (±0.03)	0.2 (±0.02)
24	Geranyl tiglate	1699	1696	1.95 (±0.61)	2.25 (±0.34)	2.28 (±0.30)
Monoterpene hydrocarbons				0.28 (±0.03)	0.20 (±0.01)	0.31 (±0.03)
Oxygenated monoterpenes				73.82 (±4.50)	71.16 (±2.89)	69.66 (±1.95)
Sesquiterpene hydrocarbons				7.88 (±1.44)	8.26 (±1.21)	7.24 (±0.29)
Oxygenated sesquiterpenes				11.33 (±2.14)	12.01 (±1.16)	13.74 (±1.17)
Total Identified				93.31	91.63	90.95

RIc: linear retention index, experimentally determined using homologous series of C_9-C_{25} alkanes.
RII: linear retention index taken from Adams (2017).
sd: standard deviation

these responses most likely related to improvements in carbon assimilation. In *Brassica napus* plants treated with seaweed extracts, a positive correlation was found between nitrogen and chlorophyll *b* content in leaves, and part of the N uptake probably was invested in photosynthesis. Improvements in photosynthetic rates were linked to the enhanced activity of Rubisco and carbonic anhydrase enzymes (23).

Seaweed extracts are a rich source of organic compounds that induce a positive response in all areas of plant metabolism. The components of the whole extract work in synergy to act on various metabolic networks

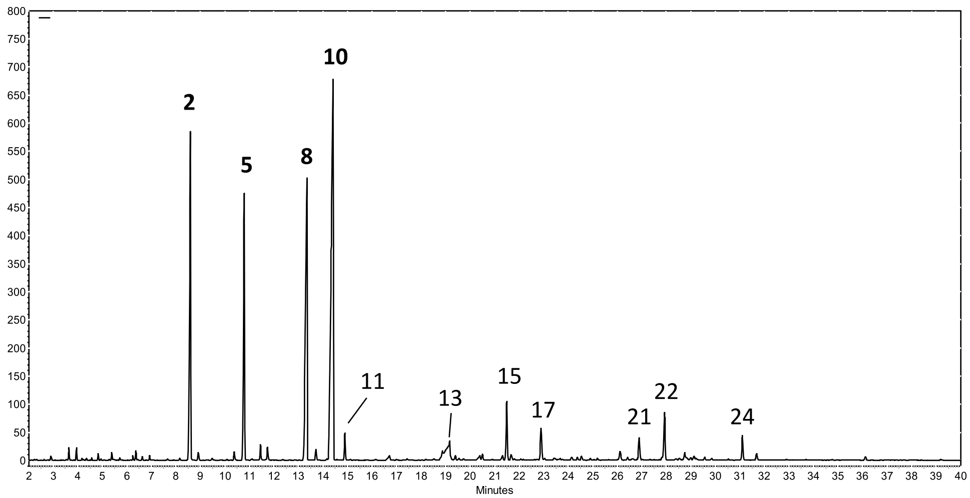

Figure 5. GC Chromatogram: Main Compounds Identified in the *Pelargonium Graveolens* Essential Oil. 2. Linalool; 5. *Iso*-Menthone; 8. Citronellol; 10. Geraniol; 11. Geranial; 13. β-Bourbonene; 15. Citronellyl Propionate; 17. Germacrene-D; 21. 2-Phenyl Ethyl Tiglate; 22. 10-*Epi*-γ-Eudesmol; 24. Geranyl Tiglate.

Table 2. Relative Proportion of Geraniol, Citronellol, Linalool and *Iso*-Menthone in *Pelargonium Graveolens* Essential Oil As a Function of *Solieria Chordalis* (SC) and *Ascophylum Nodosum* (AN) Treatments.

Treatments	Geraniol (%)	Citronellol (%)	Linalool (%)	*iso*-Menthone (%)
Control	31.13 ± 0.68 a**	16.03± 0.68^{ns}	14.13 ± 1.27 ab*	9.22 ± 0.84^{ns}
SC	27.76 ± 0.67 b	14.83 ± 0.39	16.58 ± 0.62 a	8.73 ± 0.23
AN	28.78 ± 0.56 ab	16.45 ± 0.15	13.07 ± 0.32 b	8.52 ± 0.42

Means followed by the same letters express significantly equal values by the Tukey test (n = 5).
ns non-significant, *Significant at the 5% probability level, **Significant at the 1% probability level

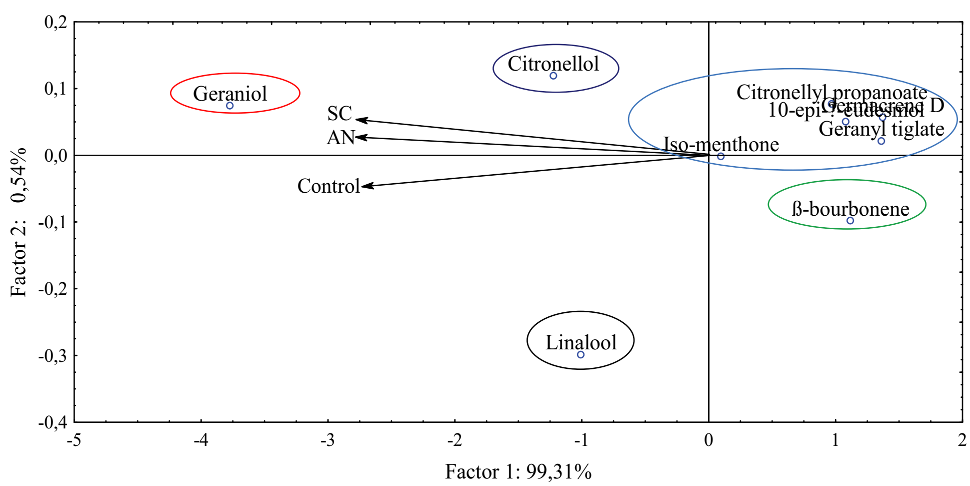

Figure 6. Analysis of Principal Components of Majoritarian Compounds Identified in *Pelargonium Graveolens* Essential Oil As a Function of *Solieria Chordalis* (SC) and *Ascophylum Nodosum* (AN) Treatments.

either independently or interactively. The major plant physiological processes positively affected by *A. nodosum* applications are nutrient absorption and assimilation (mainly sulfur and nitrogen), transport of ions and water, carbon metabolism (photosynthesis and respiration), phytohormone synthesis and physiological activity, cell division, plant development, senescence, responses to stress and antioxidant capacity (9,10). These accumulative booster effects have been shown to lead to improved plant growth, accompanied by a differential gene expression (23,24). The enhanced plant morphological and physiological performance in response to seaweed biostimulants were also observed in other aromatic species, such as mint and basil (25), coriander (26) and fennel (27) and *Cordia curassavica* (28).

In the current study, seaweeds foliar application stimulated the biosynthesis of essential oil, thereby increasing the essential oil yield (Figure 4B). In a study with algae extract from the genera *Ascophyllum*, *Sargassum* and *Laminaria*, rosemary plants also obtained greater production of leaf area and essential oil (29). Primary metabolites produced in photosynthesis and tricarboxylic acid cycle are used as precursors for the two biosynthetic pathways for terpene production in plants (7). This, in addition to the higher number of leaves, may have been one of the reasons that generated the increase in geranium essential oil production, as there was possibly an improvement in photosynthesis, demonstrated by the increase in total soluble sugars levels in plants. Furthermore, the presence of major and trace elements in seaweed extracts as well as secondary metabolites elicitors had been correlated with increased essential oil content and quality in several aromatic and medicinal plants (25). Polysaccharides (alginates, fucans, laminarin and carrageenans) are the main elicitors present in seaweed extracts (76% of the dry weight). These carbohydrates are attached to receptor proteins on plant cell membranes, leading to the activation of intracellular signaling pathways. These generally involve G proteins, protein kinases and ion channels. These signal transducers transmit the elicitor signal to downstream cascades that lead to the production of reactive oxygen species (ROS), hormonal signals and finally culminate in the expression of genes that encode defense responses of the plant (10). Essential oils are secondary metabolites related to plant defense against biotic and abiotic stresses (7). The essential oil accumulation in response to elicitation is regulated by genes and transcriptional factors involved in the plastidial methyl-erythritol-phosphate (MEP) pathway (30).

The essential oil composition matched those described in geranium plants, but there were no significant changes in oil main constituents in response of seaweed treatments, with exception of Geraniol witch was slightly reduced in SC treated plants (Table 2). However, concomitant increases in essential oil content and main constituents are a common response of aromatic plants when treated with seaweed extracts (25,27,28).

Seaweed extracts were shown to be highly effective in pot cultivation of geranium, where foliar application resulted in concomitant increases in leaf production and biosynthesis of essential oil, without modification of the essential oil composition. These discoveries emphasize the use of seaweed extracts as a tool in integrated management plans to achieve sustainability in geranium essential oil production.

5. Conclusion

Seaweed biostimulants based on *Solieria chordalis* and *Ascophylum nodosum* applied as foliar treatments are an efficient agronomical practice to maximize leaf yield and essential oil production in geranium plant. Further studies conducted at field conditions are required to support the current report.

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